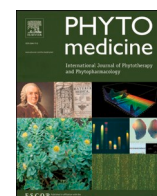




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Systematic exploration of *Astragalus membranaceus* and *Panax ginseng* as immune regulators: Insights from the comparative biological and computational analysis

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ABSTRACT

Background: Immune system plays a decisive role for defending various pathogenic microorganisms. *Astragalus membranaceus* (AM) and *Panax ginseng* (PG) are two tonic herbs used in traditional Chinese medicine (TCM) as immune booster and help to control diseases with their healthy synergistic effect on immune system.

Purpose: This study was aimed to investigate the promote effect and molecular mechanisms of AM and PG on immune system as booster and to control the target diseases using animal and computational systematic study.

Methods: Computational models including absorption, distribution, metabolism, and elimination (ADME) with weighted ensemble similarity (WES) algorithm-based models and ClueGo network analysis were used to find the potential bioactive compounds targets and pathways, which were responsible for immune regulation. Viscera index analysis, proliferation activity of splenic lymphocytes and cytotoxic activity of NK cells assays were performed to validate the effect of AM and PG on immune system of long-term administrated mice. Metabonomic study of mice plasma was conducted to investigate effect of AM and PG on the endogenous metabolic perturbations, together with correlation analysis.

Results: AM and PG simultaneously showed the ability to strengthen the immune system function including enhancement of spleen and thymus index, proliferation of splenic lymphocytes and cytotoxic activity of NK cells. Besides, the different molecular mechanisms of AM and PG on immune regulation were also investigated by analyzing the potential bioactive compounds, enzymes actions and pathways. Quercetin, formononetin and kaempferol were the main immune-related compounds in AM, while ginsenoside Ra1, ginsenoside Rh1 and kaempferol in PG. About 10 target proteins were found close to immune regulation, including acetylcholinesterase (ACHE, common target in AM and PG), sphingosine kinase 1 (SPHK1), cytidine deaminase (CDA), and Choline O-acetyltransferase (CHAT). Glycerophospholipid metabolism was regulated in both AM and PG groups. Pyrimidine metabolism and sphingolipid metabolism were considered as the special pathway in AM groups. Energy metabolism and glycerolipid metabolism were the special pathways in PG groups.

Abbreviations: ACHE, acetylcholinesterase; ADME, absorption, distribution, metabolism, elimination; AKR1B1, aldose reductase; AM, *Astragalus membranaceus*; AMP, Agilent Mass Hunter Profinder; BC, blank control; CDA, cytidine deaminase; CHAT, Choline O-acetyltransferase; Con A, concanavalin A; C-T, compound-target; CTD, comparative toxicogenomics database; DAVID, database for annotation, visualization and integrated discovery; DHCR7, 7-dehydrocholesterol reductase; DL, drug-likeness; DUOX2, dual oxidase 2; GAA, lysosomal alpha-glucosidase; GLB1, beta-galactosidase; HMDB, Human Metabolome Database; HSD11B2, corticosteroid 11-beta-dehydrogenase isozyme 2; ICR, institute of cancer research; KEGG, kyoto encyclopedia of genes and genomes; LDH, lactic dehydrogenase solution; MPP, Mass Profiler Professional; MS, Mass spectrometry; NDUFA1, acyl carrier protein, mitochondrial; NK, natural killer; NR3C1, glucocorticoid receptor; OB, oral bioavailability; OD, optical density; OPLS-DA, orthogonal partial least squares discriminant analysis; PCA, principal components analysis; PG, *Panax ginseng*; PharmGKB, pharmacogenomics knowledgebase; PPP1CC, serine/threonine-protein phosphatase PP1-gamma catalytic subunit; PYGM, glycogen phosphorylase; QC, quality control; RSD, relative standard deviation; SI, stimulation index; SPHK1, sphingosine kinase 1; SRD5A1, 3-oxo-5-alpha-steroid 4-dehydrogenase 1; SSCs, spleen suspension cells; TCM, traditional Chinese medicine; TCMS, traditional Chinese medicine systems pharmacology database; TTD, therapeutic target database; UGT2B7, UDP-glucuronosyltransferase 2B7; UPLC, Ultra Performance Liquid Chromatography; VIP, variable importance in the projection; WES, weighted ensemble similarity.

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Conclusion: A novel comprehensive molecular mechanism analysis method was established and applied to clarify the scientific connotation of AM and PG as immune regulation, with similar herbal tonic effect provided in clinical practice of TCM, which can provide a new line of research for drug development (immune booster) using AM and PG.

Introduction

Immune system is a huge and complex network with innate and acquired immunity (Castelo-Branco and Soveral, 2014). In this elaborate system, thousands of immune cells such as lymphocytes, neutrophils, macrophages and antigen presenting cells cooperate with each other and produce remarkable positive effect (Nayar et al., 2015). In this way, immune system provides continuous protection against the attack of microbial agents and maintains homeostasis together with repairing damaged tissue in human body (Zhao, 2010). If this system is broken, our body present a state of being susceptible to fatal virus, resulting in psychiatric disorders (Jones and Thomsen, 2013), cardiovascular diseases (Abboud et al., 2012), kidney diseases (Kurts et al., 2013), and many other disorders. In other words, the occurrence of various diseases is closely related to immune system. Consequently, it is necessary to develop a novel strategy to investigate the mechanisms of immune regulation and provide reference for the treatment of these diseases.

Traditional Chinese medicine (TCM) is a comprehensive medicinal system which has been potential utilization from thousands of years in clinical practice of China (Li et al., 2012; Zhao et al., 2015), providing the characteristic of multi-components and multi-targets and presenting the potent synergistic effect for treatment of various diseases with fewer side effects (Shi et al., 2014; Yang et al., 2017). Because of these incomparable advantages to modern medicine, TCM has become a global hot field in modern medical research. *Astragalus membranaceus* (AM) and *Panax ginseng* (PG) are two typical TCM tonic herbs and they are both listed in 2015 Edition of Chinese Pharmacopeia. AM has been used to treat a wide range of inflammatory (Lai et al., 2014), oxidative stress (Elabd et al., 2016), tumor (Cho and Leung, 2007), fatigue (Yeh et al., 2014), diabetes and nephropathy (Li et al., 2011), immune system (Chao et al., 2017; McCulloch et al., 2006; Mo et al., 2016; Shao et al., 2004; Wang et al., 1992; Yang et al., 2016). Pharmacological studies have also shown that PG has the efficacy of anti-cancer (Lobina et al., 2014), blood pressure regulation (Lei et al., 2012), cognitive recover (Park et al., 2015), neuroprotective (Jiang et al., 2015), anti-inflammatory (Borts et al., 2011), antioxidant (Chen and Huang, 2019; Xiong et al., 2019), immune system (Liu et al., 2011; Shin et al., 2006). Both AM and PG could improve fatigue, enhance immunity and promote recovery of body against attacked from various diseases. From thousands of years of clinical practice of TCM, people trend to use these two drugs together for tonifying Qi, their synergistic effects are thought to be occurred mainly due to presence of numerous bioactive compounds (Li et al., 2016a; Popovich and Kitts, 2004; Wan et al., 2013). AM and PG were used as immune buster for good health with state under suitable dosages, acted as a long-term medicinal drug in ancient China. These plants help to build a stable state of internal environment in mice after long-term administration. In this way, it is easy to strengthen the influence of medication on mice and amplify immune-related difference. So, long-term administration model is suitable for investigating the influence of immune regulation.

However, the synergetic effect and as immune boosting tonic in Chinese herbal formulas is still not clear, without proper description about molecular mechanisms of action, targets and signaling pathways of these two herbs. Therefore, a novel systems pharmacology approach was in urgent need to be established and applied to uncover the immune regulation mechanisms. First, the bioactive compounds with good pharmacokinetic properties in AM and PG were collected by oral bioavailability (OB) and drug-likeness (DL) screening. Secondly, multiple targets were predicted and determined by a comprehensive method

(Yu et al., 2012). And then, compounds-targets network was interpreted, and immune-relevant biological pathways were elaborated by analysis of the bioinformatics annotations of gene function and biological function. Finally, pharmacodynamic and metabonomic studies were conducted to demonstrate the reliability and validity of method using animal models. Pharmacological mechanism studies for AM and PG are usually investigated by using metabonomic strategies (Li et al., 2017a; Lin et al., 2016; Qu et al., 2016; Wu et al., 2014, 2018). Meanwhile, metabonomic research is an omics approach which uses high-throughput, high-sensitivity and high-precision method to analyze metabolites (biomarkers) *in vivo* (Lindon et al., 2006).

Presently, a comprehensive pharmacology-based approach including compounds targets network interpretation, pharmacodynamic investigation and metabonomic analysis is established. Our results provide an important evidence for insight into the molecular mechanism of AM and PG on immune regulation and offer a novel strategy for discovering new drugs from TCM.

Materials and methods

Reagents and materials

MTT cell proliferation and Cytotoxicity Detection Kit, RPMI 1640 medium (10% newborn bovine serum-10% NBS, penicillin, streptomycin) and RPMI 1640 medium (penicillin, streptomycin) were obtained from Nanjing Keygen Biotech Co. Ltd. (Nanjing, China). Con A, hank's Balanced Salt Mixture and TRIS were purchased from Beijing Solarbio Science & Technology Co. Ltd. (Beijing, China). 10% NP-40 was acquired from Beijing Leagene Biotech Co. Ltd. (Beijing, China). YAC-1 cells were obtained from Shanghai Institutes for Biological Sciences of the Chinese Academy of Sciences (Shanghai, China). Uracil was purchased from National Institute for the Control of Pharmaceutical and Biological Products (Beijing, China). Methanol, acetonitrile and formic acid were purchased from Dikma Technologies Inc., (Beijing, China). Ultrapure water was produced with a Milli-Q water purification system from Millipore (MA, USA). All other reagents used in the present study were analytical grade.

Molecular database construction

The chemical ingredients present in these herbs (AM and PG) were screened and collected from traditional Chinese medicine systems pharmacology database (TCMSP, <http://lsp.nwsuaf.edu.cn/>) (Ru et al., 2014) and pertinent literature (Li et al., 2017b; Liu et al., 2018a). Although glucosides are difficult to be absorbed after orally administration, but they can be metabolized into their deglycosylation products with the aid of intestinal microflora (Nemeth et al., 2003). So, both glucosides and its deglycosylation products are considered as potential compounds candidate for this study.

ADME-systems evaluation

To make a further investigation into potential bioactive compounds of AM and PG, *in silico* models for absorption, distribution, metabolism, elimination (ADME) are established and it is used to evaluate pharmacokinetic and pharmaceutical properties of compounds obtained from AM and PG. These chosen models are OB and DL. The description of these two models was provided in section OB screening and DL screening

OB screening

OB is a key pharmacodynamic parameter which reflect the delivery situation of the orally administered drug in greater circulation (van der Graaf and Benson, 2011; Xu et al., 2012). In this study, OB screening test was performed using *in silico* model OBioavail1.1 (Xu et al., 2012). Compounds with OB values $\geq 30\%$ were obtained as the candidate molecules for further analysis.

DL screening

To obtain satisfied potential molecules, DL model was used to remove the compounds with poor chemical and pharmacological properties in AM and PG. The DL values are got by calculating the Tanimoto Coefficient-based Similarity (Willett et al., 1998) between aimed herbal compounds and drugs in the DrugBank database (<http://www.drugbank.ca/>) (Wishart et al., 2018). The DL prediction approach was calculated as follows:

$$T(A, B) = \frac{A \cdot B}{|A|^2 + |B|^2 - A \cdot B}$$

Where, A: molecular description of herbs screened compounds, B: average molecular DL index description of all compounds in DrugBank database. In this work, herb compounds with $DL \geq 0.18$ are selected as the candidate molecules for further analysis. The compounds that screened by the OB and DL models were regarded as candidate compound. However, some compounds such as daidzein (Wei et al., 2012a), astragaloside I (Ying-Yuan et al., 2017), astragaloside II (Wan et al., 2013), astragaloside IV (Li et al., 2016a), gamma-aminobutyric acid (Tian et al., 2004), ginsenoside-Rh1 (Jeong et al., 2013), 20(S)-ginsenoside-Rg2 (Ma and Yu, 1993), 20(s)-protopanaxadiol (Popovich and Kitts, 2004), 20-(S)-Ginsenoside-Rg3 (Park et al., 2008) compounds were artificially omitted according to our screening rules. Besides, some derivatives or analogs from reported bioactive compounds are also taken into candidate compounds list.

Drug targeting analysis

It is necessary to build the compound-target (C-T) interaction network (Rix and Superti-Furga, 2009) to clarify the immunomodulatory mechanism of AM and PG. In this work, we successfully performed the weighted ensemble similarity (WES) algorithm model to predict the interactions between direct targets and bioactive compounds based on a large-scale of drug-target relationships (Zheng et al., 2015). The matched targets were uploaded to Uniprot (<http://www.uniprot.org/>) for normalization. In addition, multiple target databases were used to screen immune-related information together, including DAVID (<https://david.ncifcrf.gov/summary.jsp>), TTD (<http://database.idrb.cqu.edu.cn/TTD/>), CTD (<http://ctdbase.org/>), PharmGKB, (<https://www.pharmgkb.org/>) and KEGG (<http://www.kegg.jp/>).

Network construction

In order to elucidate the synergism mechanisms of AM for PG at a network level, we established a visualized network, that is C-T network. All active compounds from AM and PG, together with their potential targets were used to generate a complex interactions graph between drug and target, which presented the situation of direct connections among compounds and targets. This network was created by Cytoscape version 3.7.0, which is a website with network data integration, analysis, and visualization program (Smoot et al., 2011). In this C-T network, compounds and targets were represented by nodes in different color, and an edge was used to represent the interaction or relatedness between compound and target. Besides, crucial topological parameter named degree was used to evaluate the importance of the node in a network

(Azuaje et al., 2011). All these topological properties were analyzed by the Network Analysis function of Cytoscape. Pathway analysis is another way to clarify the therapeutic mechanism of AM and PG. In this way, the information of pathways derived from targets was analyzed by KEGG and DAVID. Besides, molecular function, reactome pathway, biological process and immune system process were also analyzed by ClueGO function of cytoscape software.

Sample collection and animal experiments

Eighty-four mice ($20 \text{ g} \pm 0.02 \text{ g}$) of institute of cancer research (ICR), half male and half female, were randomly divided into blank control (BC) group, low-dose AM group (LAM, 5 g/kg), moderate-dose AM group (MAM, 25 g/kg), high-dose AM group (HAM, 50 g/kg), low-dose PG group (LPG, 1.5 g/kg), moderate-dose PG group (MPG, 7.5 g/kg) and high-dose PG group (HPG, 15 g/kg). Before drug administration, the mice were acclimatized for 7 days in a rearing environment with 12 h light-dark cycle (light on from 6:00 to 18:00 h) at ambient temperature ($22\text{--}24^\circ\text{C}$) with 60% relative humidity. Mice were fasted for 12 h with free access to water. They were fed continuously for 30 days by intra-gastric administration of AM or PG. After administration, 1 ml serial blood sample was collected by eyeballs extraction into heparin-containing tubes and centrifuged at 3000 rpm for 15 min. Plasma was separated and stored at -20°C until analysis. All experimental procedures were approved by the Institutional Animal Care Committee of China Academy of Chinese Medical Science (2017–051).

Splenic lymphocyte proliferation assay

Mouse spleen was removed, grinded and passed through a stainless mesh (200 Φ) by aseptic operation. The final concentration of spleen suspension cells (SSCs) was adjusted to $3 \times 10^6/\text{ml}$ by RPMI1640 medium containing 10% NBS (ratios of living cells was 96–98% by trypan blue exclusion). SSCs (1 ml per well) proliferation were stimulated by Con A (final concentration 3.0 $\mu\text{g}/\text{ml}$) in 24 well cell culture cluster and then cultured at 37°C in 5% CO_2 cultivating box for 68 h. After 68 h incubation and then centrifuged at 1000 rpm/min for 10 min. The supernatant was discarded and the same volume of RPMI 1640 medium without 10% NBS was added into each well. Subsequently, MTT solution (5 mg/ml) was added into the wells with a volume of 100 μl per well, and intermixed into uniformity softly, put it into 5% CO_2 cultivating box again. After another 4 h incubation and then centrifuged at 4200 rpm for 10 min. 500 μl supernatant was discarded and 800 μl dimethyl sulfoxide was added into each well. The absorbance was measured at 570 nm by Enzyme Mark Instrument, and the average optical density (OD) value was calculated. The stimulation index (SI) was calculated according to the following formula: $SI = \frac{\text{OD}_{\text{Con A stimulated lymphocyte proliferation}} - \text{OD}_{\text{spontaneous lymphocyte proliferation without Con A}}}{\text{OD}_{\text{spontaneous lymphocyte proliferation without Con A}}}$. SI was used to assess the difference of proliferation ability between treatment groups and BC group.

Cytotoxic activity of natural killer cells assay

YAC-1 cells were used as target cells and washed by Hank's solution three times in pre-experiment. Finally, the concentration of target cells was adjusted to $4 \times 10^5/\text{ml}$ by RPMI1640 medium containing 10% NBS (ratios of living cells was 98% by trypan blue exclusion). SSCs were collected by the same method in splenic lymphocyte proliferation assay. After speed centrifugation at 1000 rpm/min for 10 min, 500 μl sterile water was added to splitting erythrocyte for 20 s and then 8.0 ml hank's solution was added. Finally, the concentration of cells was adjusted to $2 \times 10^7/\text{ml}$ by RPMI1640 medium containing 10% NBS (ratios of living cells was 97% by trypan blue exclusion).

SCCs without erythrocyte were regarded as effector cells. Equivalent amount (100 μl) of RPMI1640 medium, 1% NP40 solution and effector cells were added into the wells with 100 μl target cells in 96 well cell culture cluster, respectively. At 50:1 effector-target ratio, the lytic

activity of natural killer (NK) cell was investigated. Three independent experiments ran in triplicate. After 4 incubation in 5% CO₂ cultivating box, centrifuged at 1500 rpm/min for 10 min and 100 µl LDH was added to the supernatant. LDH contained lithium lactate (5×10^{-2} mol/l), iodinitrotetrazolium (INT, 6.6×10^{-4} mol/l), phenazine methosulfate (PMS, 2.8×10^{-4} mol/l) and nicotinamide adenine dinucleotide (NAD, 1.3×10^{-3} mol/l). The reaction was stopped by 1 mol/l HCl. The absorbance was measured at 490 nm by Enzyme Mark Instrument, and the average OD value was calculated; NK activity = (experimental - spontaneous release) / (maximum - spontaneous release) * 100%

Viscera index analysis

Spleen and thymus of mice were stripped under sterile environment, weighted and calculated after the last administration. Spleen index was calculated by spleen weight (mg) divided / weight (g). Thymus index was calculated by thymus weight (mg) / body weight (g) as described in previous method.

Sample preparation for metabolite profiling

Plasma samples were thawed at room temperature and 100 µl was transferred into an Eppendorf tube, 300 µl of ice-cold acetonitrile was added to precipitate the plasma proteins. Samples were then vortexed for 1 min and stored in refrigerated (4 °C) condition for 3 h. After that, centrifuged at 16,000 rpm for 15 min (4 °C) and the supernatant was transferred and evaporated to dryness under a vacuum centrifugal concentrator. The residue was reconstituted in 200 µl of 15% acetonitrile-water, and a 1 µl aliquot was injected for analysis.

Validation of the metabonomic method

Reproducibility and precision of method were investigated. Relative standard deviation (RSD) of detected retention time, mass-to-charge ratio (m/z), peak height and peak intensity were investigated. Typical pooled samples were used as analysis object for quality control (QC). 72 pooled plasma samples were mixed together of equal volumes (10 µl) and prepared as sample preparation method.

LC-MS and LC-MS/MS platforms for metabolite profiling

Ultra-Performance Liquid Chromatography (UPLC) analysis was conducted on a 1290 UPLC system (Agilent Technologies, Santa Clara, CA, USA). Injection volume was 1.0 µl and the flow rate of the mobile phase was 0.40 ml/min. Separation of analytes was performed on a reversed phase Agilent Zorbax XDB-C₁₈ column (2.1 mm × 50 mm, particle size 3.5 µm) with the column oven set to 35 °C. The elution gradient was eluted from the column under a gradient of 0.1% v/v formic acid in water (A) and acetonitrile (B) and optimized separation was obtained according to a linear gradient (0–1.5 min, 5% B; 1.5–3 min, 5–50% B; 3–16 min, 50–95% B; 16–21 min, 95% B, returned to 5% B for 5 min). Acetonitrile as a blank solution was run every four samples.

Mass spectrometry (MS) was performed by using a time-of-flight mass spectrometer (Q-TOF-MS/MS, 6538 UHD Accurate Mass Q-TOF; Agilent Technologies) equipped with an electrospray ionization source, operated in positive-ion and negative-ion modes. MS conditions: Capillary voltage, 4000 V (+) and 3500 V (-); Gas temperature, 350 °C; Flow rate of drying gas (N₂), 10 l/min; Nebulizer pressure, 35 psi; Skimmer voltage, 50 V; Octopole radiofrequency, 150 V; Fragmentor voltage, 150 V; Mass range, 30–1100 Da. The data were collected in profile mode using the internal reference mass for accurate, automated mass calibration correction during LC/MS runs. Purine (m/z 121.050873) and HP-0921 (m/z 922.009798) were used as internal reference masses in positive ion mode, and in negative ion mode were TFA anion (m/z 112.985587) and HP-0921 (m/z 1033.988109). The reference nebulizer was set at 1 psi.

Data processing

The collected raw data were extracted and analyzed by the function of Molecular feature extractor within Mass Hunter Qualitative Analysis (Agilent Technologies). The retention time, m/z value, and the volume for each compound were investigated. The data were converted into the .cef format by Agilent Mass Hunter Profinder (AMP, Agilent Technologies) for targeted recursive feature finding, for peak detection and alignment. Further, the .cef data were imported into Mass Profiler Professional (MPP, Agilent Technologies) for differential analysis. To compare the metabolite profiles among BC, AM and PG groups, the Student's *t*-test and analysis of variance were conducted. Compounds with the value of $p < 0.05$ and fold change ≥ 2.0 were selected as preliminary potential biomarkers. Then, the biomarker formulations were generated and queried using the Agilent METLIN Compound Database (assistant software package in MPP).

Principal components analysis (PCA) and orthogonal partial least squares discriminant analysis (OPLS-DA) models were performed by SIMCA-P 14.1 (Umetrics, Umea, Sweden). Data were pareto scaled. Variable importance in the projection (VIP) values was taken as key point for peak selection. Student's *t*-test was used to measure the significance of each metabolite. Metabolites of VIP value larger than 1.0 ($VIP > 1.0$, $p < 0.05$) was considered as characteristic metabolites. The typical fragmentation patterns of selected biomarkers were analyzed based on the MS/MS data information. Besides, the metabolic pathway analysis of proposed biomarkers was analyzed by MetaboAnalyst 3.0 with KEGG and Human Metabolome Database (HMDB, <http://www.hmdb.ca/>). Based on the obtained pathway information, a cross-over analysis of identified biomarkers was analyzed comprehensively. Correlation analysis was conducted by using SPSS 24.0 for Windows (IBM, Armonk, NY, USA). Pearson's correlation coefficient was used to investigate correlations between the four pharmacodynamics indexes and identified biomarkers, to find out immune-related biomarkers.

Results

Candidate compounds screening

Oral administration is a common administration method, where the drugs are taken through the digestive tract and reached their cellular targets in different parts of the body via the blood circulation. The drugs play a systemic effect for patients in treating various diseases in TCM therapy (Chen et al., 2017). Good pharmacokinetic properties are helpful to deliver bioactive drugs to systemic circulation. Thus, OB screening and DL evaluation models were chosen to determine the potential bioactive compounds in AM and PG. Based on this method, we did a screening work for potential bioactive compounds. First, a total of 277 compounds were collected from AM and PG based on TCMSP database. The bioactive compounds were screened out by ADME properties. So, 20 candidate chemical ingredients were obtained from AM and 17 potential ingredients were found in PG according to screening rules with OB $\geq 30\%$, DL ≥ 0.18 . Besides, we also found 15 and 47 molecules in AM and PG, respectively. All of them presented low OB ($< 30\%$) or DL (< 0.18) abilities, but they were regarded as the main abundant constituents or the most bioactive pharmacological in previous studies (Geng et al., 2013; Li et al., 2016; Liu et al., 2018b; Shi et al., 2015). Finally, 35 ingredients in AM and 64 ingredients in PG were obtained for further studies, including target prediction and network analysis, and only a common compound was found, that was kaempferol (Table 1). The obtained results showed much differences in chemical profiles of AM and PG. These various kinds of potential bioactive compounds may play multiple biological effects *in vivo*. A deeper investigation into chemical structure, all these candidate compounds could be classified as five groupies: triterpenes (55.56%), flavonoids (18.18%), polyacetylenes (6.06%), alkaloids (5.05%) and lignans (4.04%) as showed in Fig. 1A. However, the compounds which could not classified

Table 1

Chemical information of active compounds and their network parameters in AM and PG.

NO.	Compounds	InChI Key	OB (%)	DL	Degree	Type
M01	(24S)-24-Propylcholesta-5-ene-3 β -ol	KLEXDBGYSOIRE-UIFQYPGES-A-N	36.23	0.78	3	Triterpenes
M02	Quercetin	REFJWTPEDVJJY-UHFFFAOYSA-N	46.43	0.28	34	Flavonoids
M03	Betulinic Acid	QGJZLNKBHJESQX-FZFNOLFKSA-N	55.38	0.78	4	Triterpenes
M04	Kumatakenin	BJBUTJQYZDYRMJ-UHFFFAOYSA-N	50.83	0.29	20	Flavonoids
M05	Hederagenin	PGOYMURMZNDHNS-MYPRUECHSA-N	36.91	0.75	31	Triterpenes
M06	Isorhamnetin	IZQSVPOUDKVDZ-UHFFFAOYSA-N	49.60	0.31	31	Flavonoids
M07	3,9-di-O-methylinissolin	RFFNFQZKHNKPO-BBRMVZONSA-N	53.74	0.48	33	Lignans
M08	5'-hydroxyisomuronulatol-2',5'-di-O-glucoside	SRVGYVIWVOOXQO-FQRJZKGRSA-N	41.72	0.69	19	Flavonoids*
M09	7-O-methylisomuronulatol	BLHQCJBSTMDZQA-LBPRGKRZSA-N	74.69	0.30	45	Flavonoids
M10	9,10-dimethoxypterocarpan-3-O- β -D-glucoside	PCIXSTFFMHVOMF-PBGSHFJYSA-N	36.74	0.92	5	Flavonoids*
M11	Astrapterocarpan	UOVGCLXUTLXAE-WFASDCNBSA-N	64.26	0.42	32	Flavonoids
M12	Bifendate	JMZOMFYRADAWOG-UHFFFAOYSA-N	31.10	0.67	13	Others
M13	Formononetin	HKQYGTCTHOMP-UHFFFAOYSA-N	69.67	0.21	30	Flavonoids
M14	Isoflavanone	JNSVNRWHSLLCBG-LLVKDONJSA-N	109.99	0.30	9	Flavonoids
M15	Calycosin	ZZAJQOPSWVWMBI-UHFFFAOYSA-N	47.75	0.24	23	Flavonoids
M16	Kaempferol	IYRMWYMSZSQPKC-UHFFFAOYSA-N	41.88	0.24	32	Flavonoids
M17	Folic Acid	OVBPILUPVIDEAO-LBPRGKRZSA-N	68.96	0.71	2	Others
M18	(3r)-7,2'-dihydroxy-3',4'-dimethoxyisoflavan	NQRBAPDEZYMKFL-NSHDSACASA-N	67.67	0.26	10	Flavonoids
M19	Isomuronulatol-7,2'-di-O-glucoside	NHOPAJCVMIDGBN-MEPKZADGSA-N	49.28	0.62	1	Flavonoids*
M20	1,7-Dihydroxy-3,9-dimethoxy pterocarpene	RVGSZUMTFIOERY-UHFFFAOYSA-N	39.05	0.48	16	Flavonoids
M21	Gamma-aminobutyric acid	BTCSSZJGUNDROE-UHFFFAOYSA-N	24.09	0.01	11	Others
M22	Daidzein	ZQSJLRDFPHDXIC-UHFFFAOYSA-N	19.44	0.19	18	Flavonoids
M23	(+)-Syringaresinol	KOWMJRJXZMEZLD-HCIHMXRSSA-N	3.29	0.72	18	Lignans
M24	Astragaloside I	GBPDDYORNLQZID-ZHIISFHXA-N	46.79	0.11	26	Triterpenes*
M25	Astragaloside I _{qt}	HAUCZONIQLZEEW-SQLOWIESA-N	12.34	0.20	27	Triterpenes*
M26	Astragaloside II	AYWNHWGQTMQCIV-IRBWRUHMSA-N	46.06	0.13	25	Triterpenes*
M27	Astragaloside II _{qt}	KILZJJKCZIGLEJ-QOICHAAMSA-N	11.55	0.25	26	Triterpenes*
M28	Astragaloside III	FVFSMBDVZVUETN-ADMPTBELSA-N	31.83	0.10	25	Triterpenes*
M29	Astragaloside III _{qt}	VXHVFQYSSFKAR-ZNDWJNNSA-N	5.35	0.32	26	Triterpenes*
M30	Astragaloside IV	QMNWISYXSJWHRY-IZPZSTDWSA-N	22.50	0.15	23	Triterpenes*
M31	Astragaloside IV _{qt}	VXHVFQYSSFKAR-VOVRKYFDSA-N	7.07	0.32	24	Triterpenes*
M32	Astrasierversianin XV	KQSZHSKJNNXOTD-FWLWVOKOSA-N	11.19	0.07	22	Triterpenes*
M33	Alpha-L-Rhamnose	SHZGCJCMOBCMKK-HGVZOGFYSA-N	50.50	0.04	14	Others
M34	Betaine	KWUHFFTVRNATP-UHFFFAOYSA-O	40.92	0.01	29	Others
M35	Isoferulic acid	QURCVMIKCOAJU-HWKANZROSA-N	50.83	0.06	16	Others
M36	Beta-sitosterol	KZJWDPNRJALLNS-VJSFXLFLSA-N	36.91	0.75	44	Triterpenes
M37	Kaempferol	IYRMWYMSZSQPKC-UHFFFAOYSA-N	41.88	0.24	32	Flavonoids
M38	Stigmasterol	HCXVJBMSMIARIN-PHZDYDNGSA-N	43.83	0.76	42	Flavonoids
M39	Fumarine	GPTFURBXHJWNHR-UHFFFAOYSA-N	59.26	0.83	41	Alkaloids
M40	Diop	FBVWWBDCUKLFFN-UHFFFAOYSA-N	43.59	0.39	7	Others
M41	Inermin	HUKSJTUUSUGIDC-BDJLRTHQSA-N	65.83	0.54	30	Lignans
M42	Aponorhyscine	UPWMWFSEBOFTNA-UHFFFAOYSA-N	66.65	0.22	8	Alkaloids
M43	Deoxyharringtonine	WRCBXHDQHPVUHW-CDRKEARJSA-N	39.27	0.81	2	Alkaloids
M44	Dianthramine	SVZLTRRSGWXPBL-UHFFFAOYSA-N	40.45	0.20	5	Alkaloids
M45	Arachidonate	YZXBAPSDXZZRQB-CGRWFSSPSA-N	45.57	0.20	9	Fatty Acids
M46	Frutinone A	RFWULRHBGYKEEZ-UHFFFAOYSA-N	65.90	0.34	26	Flavonoids
M47	Ginsenoside Rh ₄ _{qt}	JKPOYAJYRYOGBN-ARWCPIWWSA-N	31.11	0.78	5	Triterpenes
M48	Girinibin	GAEQWKVGMHUUKO-UHFFFAOYSA-N	61.22	0.31	22	Alkaloids
M49	Malkangunin	DTMIMKTZETWDJV-DUUKBJRLSA-N	57.71	0.63	1	Others
M50	Panaxadiol	PVLHOJXLNBFHDX-BMTJSWHCSA-N	33.09	0.79	2	Triterpenes
M51	Suchilactone	GVNUFBXIXQNOCF-CRNMQVKPSA-N	57.52	0.56	29	Lignans
M52	Alexandrin _{qt}	KZJWDPNRJALLNS-HEEOUYHISA-N	36.91	0.75	3	Triterpenes
M53	Panaxynol	UGJAEDFOKNAMQD-QXPXGGMISA-N	42.44	0.10	3	Polyacetylenes
M54	Ginsenoside Ro _{qt}	MIJYXULNPSFWEK-CGXHKNNHBSA-N	17.62	0.76	3	Triterpenes
M55	Ginsenoside Rh ₁	RAQNTCRNSXYLAH-UHFFFAOYSA-N	3.86	0.57	24	Triterpenes*
M56	Ginsenoside Rh ₁ _{qt}	SHCBCKBYTHZQZG-AJZBJTNWSA-N	20.13	0.78	2	Triterpenes
M57	20(S)-Ginsenoside Rg ₂	AGBCLJAHARWNLA-CBWWKSIWSA-N	10.09	0.26	25	Triterpenes*
M58	20(S)-Protopanaxadiol	PYXFVCFISTUSOO-VUFVRDRTSA-N	29.69	0.77	2	Triterpenes
M59	20(R)-Ginsenoside Rg ₂	AGBCLJAHARWNLA-LVJGBEOWSA-N	10.09	0.26	25	Triterpenes*
M60	20(S)-Ginsenoside Rg ₃	RWXIFXNRCLMQCD-LYTHEZCASA-N	13.69	0.22	30	Triterpenes*
M61	20(S)-Ginsenoside Rg ₃ _{qt}	PYXFVCFISTUSOO-DCDFIXCCSA-N	29.69	0.77	22	Triterpenes
M62	6'-Malonylginsenoside Rd ₁	NTYAVUNEPXGZQJ-JPOZCNCBSA-N	6.70	0.03	22	Triterpenes*
M63	6'-Malonylginsenoside Rd ₁ _{qt1}	PYXFVCFISTUSOO-BFYBINCJSA-N	29.69	0.77	22	Triterpenes
M64	Ginsenoside La	CYUTWBUOPDPXJL-FHZZWBSINSA-N	17.74	0.14	28	Triterpenes*
M65	Ginsenoside La _{qt}	XSFXFJMBHIGYOZ-KMPFTHCMSA-N	15.70	0.78	2	Triterpenes
M66	Ginsenoside Ro	NFZYDZXHKFHPGA-UPULUQVQSA-N	1.98	0.05	29	Triterpenes*
M67	Ginsenoside Ra ₀	LDIAQNKCRXZCD-UJQRMWHLSA-N	7.30	0.01	23	Triterpenes*
M68	Gynposide V _{qt}	PYXFVCFISTUSOO-XPRAATSESA-N	29.69	0.77	2	Triterpenes*
M69	Ginsenoside Ra ₁	KVMXBSSOCCPAOR-KHMVMXOBSA-N	7.09	0.02	25	Triterpenes*
M70	Ginsenoside Ra ₂	UEBIBJSHWIZNCA-MUXWUWCSSA-N	7.62	0.02	20	Triterpenes*
M71	Ginsenoside Ra ₃	QUNSGRLNZDSQJC-AYBHOFBPSA-N	7.19	0.02	21	Triterpenes*
M72	Ginsenoside Rb ₁	GZYPWOGIYAIIPV-MPBJDOIPSA-N	6.24	0.04	21	Triterpenes*
M73	Ginsenoside Rb ₂	NODILNFGTFIURN-GAAOXUKQSA-N	6.02	0.04	24	Triterpenes*
M74	Gypenoside LXIX	NODILNFGTFIURN-FCUDRQNNSA-N	7.73	0.04	27	Triterpenes*

(continued on next page)

Table 1 (continued)

NO.	Compounds	InChI Key	OB (%)	DL	Degree	Type
M75	Ginsenoside Rc	JDCPEKQWFDWQLI-ZCKGZTPXSA-N	8.16	0.04	22	Triterpenes*
M76	Ginsenoside Re	PWAOJDMFUQOKB-WCZZMFLVSA-N	4.27	0.12	21	Triterpenes*
M77	Ginsenoside Rg3	RWXIFXNRCLMQCD-DRJVPZJKSA-N	17.75	0.22	29	Triterpenes*
M78	Ginsenoside Rg3_qt	PYXFVCFISTUSOO-HHYRSGPWSA-N	29.69	0.77	2	Triterpenes
M79	Ginsenoside Rh3_qt	OURRZLJECNEIMI-MQCBTYGISA-N	13.09	0.76	4	Triterpenes
M80	Ginsenoside Rh4	OZTXYFOXQFKYRP-IKDRHPKYSA-N	5.22	0.60	26	Triterpenes*
M81	Ginsenoside Rs1	HAJTYGKLQDXTPO-UHFFFAOYSA-N	6.27	0.04	22	Triterpenes*
M82	Ginsenoside Rs2	YLMBQJRKOKVUCP-UHFFFAOYSA-N	8.14	0.04	22	Triterpenes*
M83	Ginsenoside A	FTXZFRHQNZNH-UHFFFAOYSA-N	66.22	0.13	1	Polyacetylenes
M84	Ginsenoside B	MORPELUWUARUFU-ULQDDVLXSA-N	39.79	0.13	2	Polyacetylenes
M85	Ginsenoside C	WNVDKDQMWFSCL-ULQDDVLXSA-N	43.38	0.13	18	Polyacetylenes
M86	Ginsenoside D	WDZQEROINMBCOK-YESZJQIVSA-N	19.32	0.13	7	Polyacetylenes
M87	Ginsenoside E	WIONCQLWGYLTME-UHFFFAOYSA-N	36.53	0.13	1	Polyacetylenes
M88	Malonylginsenoside Rc	UOFHLCPPZXRFL-RJBDEUDKSA-N	7.84	0.03	20	Triterpenes*
M89	Malonylginsenoside Rc_qt1	PYXFVCFISTUSOO-SPPDXTJESA-N	29.69	0.77	22	Triterpenes*
M90	Malonylginsenoside Rd	OSXWNRAKZUNVDR-HUOLQKEOSA-N	8.84	0.07	25	Triterpenes*
M91	Malonylginsenoside Rd_qt	PYXFVCFISTUSOO-QQSXEGFZSA-N	29.69	0.77	2	Triterpenes
M92	beta-D-Mannose	WQZGKKKJLJFFOK-RWOPYEJCSA-N	43.04	0.04	20	Others
M93	Nonacosanediol-6,8	IAABKQJHJKGRFAU-MNOVXSKEA-N	17.79	0.03	20	Others
M94	Notoginsenoside R2	FNIRVWPHRMMRQI-PGOMJGFYSA-N	17.74	0.28	24	Triterpenes*
M95	Notoginsenoside R2_qt	SHCBCKBYTHZQZG-AELBPGHMSA-N	20.13	0.78	2	Triterpenes
M96	Notoginsenoside R6	YPUHYSBFIMWSEC-UHFFFAOYSA-N	4.70	0.12	20	Triterpenes*
M97	Ginsenoside Rg5	VNVVACHTDGQAH-A-DONNJNOYSA-N	6.15	0.23	34	Triterpenes*
M98	Panaxatriol	QFJUYYMMIBFBOJY-UXZRXANASA-N	15.42	0.79	2	Triterpenes
M99	Ginsenoside rf	UZIOUZHBUYLDHW-XUBRWAZSA-N	17.74	0.24	22	Triterpenes*

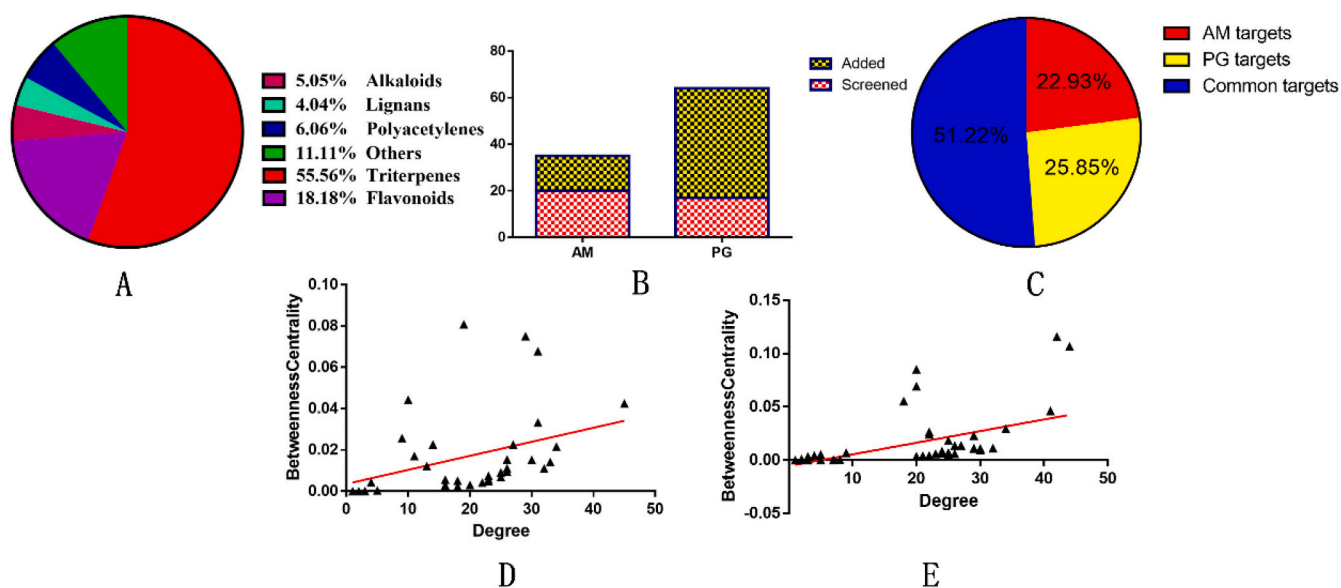


Fig. 1. (A) Distribution of bioactive compounds according to their chemical criteria. (B) Number of bioactive compounds in AM and PG. (C). Percentage of common targets and non-common targets in AM and PG. (D and E) The correlation plots of the betweenness centrality and degree for compounds and target sets, in AM and PG, respectively.

into above-mentioned groups were regarded as others (Fig. 1B).

Target predication and analysis

In TCM practice, various diseases could be effectively prevented through by one or more drugs, on the base of synergistic effects of multiple compounds binding to biological targets *in vivo* (Hao and Xiao, 2014). Therefore, targets investigation is conducted, and it contribute toward understanding the therapeutic mechanism. Finally, 205 potential targets were obtained by WES algorithm model, and connections between them were 1824. Among them, 105 targets were common targets between AM and PG. Besides, 47 and 53 targets were special target in AM and PG, respectively. The rate of common targets reached 51.22% (Fig. 1C). Even though there was much differences in chemical compounds level between these two herbs (AM and PG), the targets they

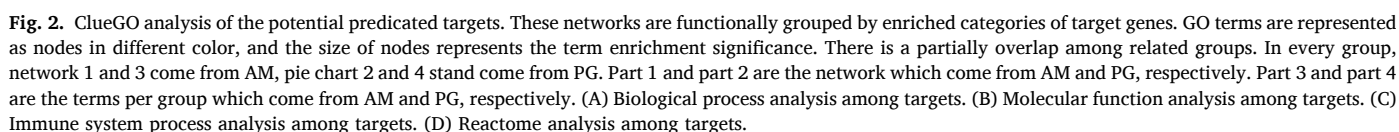
bind showed simultaneous similarity, indicating similar targets regulation to exert synergistic effects.

Compound-target network and analysis

Multicomponent therapeutics were the essential character of TCM, and typical bioactive compounds of herbs could interact its targets *in vivo*. After compounds screening and target predication, the interactions among compounds and their related targets were investigated systematically by constructing the C-T network. This network was built on 304 nodes (99 potential compounds and 205 targets) and 1824 edges. The results showed that the average degree number of targets in AM is 20.62 per compound, and 17.22 per compound in PG, indicating the highly multi-target properties and poly pharmacology in AM and PG. In these compounds, 59 were found with high degree distributions and the

was speculated that flavonoids are the main bioactive compounds in AM and triterpenes were in PG. Furthermore, an investigation was conducted at the relationships between betweenness centrality and degree (Fig. 1D and E). There was a strong correlation among them, and these calculated *p* values were both less than 0.05, indicating a linear correlation coefficient in betweenness centrality and degree.

Among the studied compounds with high degree, quercetin (Li et al., 2016b), formononetin (Mansoori et al., 2016), beta-sitosterol (Alappat et al., 2010), stigmaterol (Antwi et al., 2017), 20(S)-ginsenoside Rg3 (Wei et al., 2012b) were reported to be the bioactive ingredients in regulating the immune system. Kaempferol could also effectively regulate immune responses by exhibiting an immunosuppressive effect on dendritic cells (Lin et al., 2011). Thus, compounds from AM and PG



mainly showed the function of immune regulation. Besides, glucocorticoid receptor (NR3C1) was the target with highest degree, connections with 53.53% (53 of 99) of the compounds. Receptor for glucocorticoids could binds to glucocorticoid response elements such as nuclear and mitochondrial DNA. It could also modulate the other transcription factors. In these ways, inflammatory responses, biological cell proliferation and differentiation were regulated *in vivo*. Furthermore, acyl carrier protein, mitochondrial (NDUFAB1, degree 40), 7-dehydrocholesterol reductase (DHCR7, degree 38), dual oxidase 2 (DUOX2, degree 38), corticosteroid 11-beta-dehydrogenase isozyme 2 (HSD11B2, degree 38), serine/threonine-protein phosphatase PP1-gamma catalytic subunit (PPP1CC, degree 38), 3-oxo-5-alpha-steroid 4-dehydrogenase 1 (SRD5A1, degree 38) were the most highly connected targets.

To get a deeper understanding of the action mechanism of 205 potential targets, ClueGO analysis was applied to elucidate the biological process happened *in vivo* (Bindea et al., 2009)

The results were shown in Fig. 2 and this figure could be divided into four parts, such as biological process (A), molecular functions (B), immune system processes (C) and the reactome pathway (D). The Fig. 2A showed that the common biological processes in AM and PG were regulation of neurotransmitter levels, blood circulation, response to hypoxia, response to oxygen-containing compound, G protein-coupled receptor signaling pathway and organic hydroxy compound metabolic process, which indicated that blood system and nervous system could be the main biological reaction places. The common molecular functions of AM and PG were transmembrane signaling receptor activity, nuclear receptor activity, neurotransmitter receptor activity, peptide binding, zinc ion binding and oxidoreductase activity (Fig. 2B). It could be seen that large number of targets were related to neurotransmitter receptor activity and oxidoreductase activity. The results of immune system processes analysis (Fig. 2C) showed that macrophage chemotaxis and response to interferon-gamma were the common process associated with targets in AM and PG. Besides, positive regulation of antimicrobial peptide production, hematopoietic progenitor cell differentiation and T cell costimulation were found in AM. Toll-like receptor 4 signaling pathway and regulation of lymphocyte migration were in PG. Finally,

Fig. 2D showed that interleukin-4 and interleukin-13 signaling, nuclear receptor transcription pathway, transmission across chemical synapses and nitric oxide stimulates guanylate cyclase were the intimately related reactome of targets in AM and PG. Besides, neurotransmitter clearance, platelet aggregation, VEGFA-VEGFR2 pathway, fatty acid metabolism and sphingolipid metabolism also played important part in AM, or PG-mediated biochemical reaction in living organisms. Besides, the possible pathways were also analyzed by KEGG with a web-accessible program, named David. To clarify drugs characteristic effects, the sites of its action must be answered clearly. Thus, a clear theoretical evidence was obtained in this experiment and found that the flavonoids and triterpenes were most active compounds of immune regulation *in vivo*, binding to most protein targets. Besides, with the decipherment of targets, its sites of action were also found to related to the blood micro-environment, nervous and immune system. From our results, targets with multiple biological functions shared by active ingredients indicated that AM and PG have the effects of immune regulation.

Effect of AM and PG on mice weight and viscera index

There was no significant difference in the body weights of mice between treatment groups and BC group in Fig. 3A. Besides, no significant difference was also found in thymus index in Fig. 3B. The index numbers of spleen of MPG, HPG, LAM and MAM groups were significantly higher than BC group ($p < 0.05$) in Fig. 3C. It showed that long-term administration of AM or PG contributed to strengthen the spleen index of mice, compared to the BC group.

Effect of AM and PG on the proliferation of splenic lymphocytes

The SI in treatment groups with AM and PG all showed a significant increase as compared with that in BC group in Fig. 3D. It indicated that AM and PG improved adaptive immunity in mediating proliferation of splenic lymphocytes.

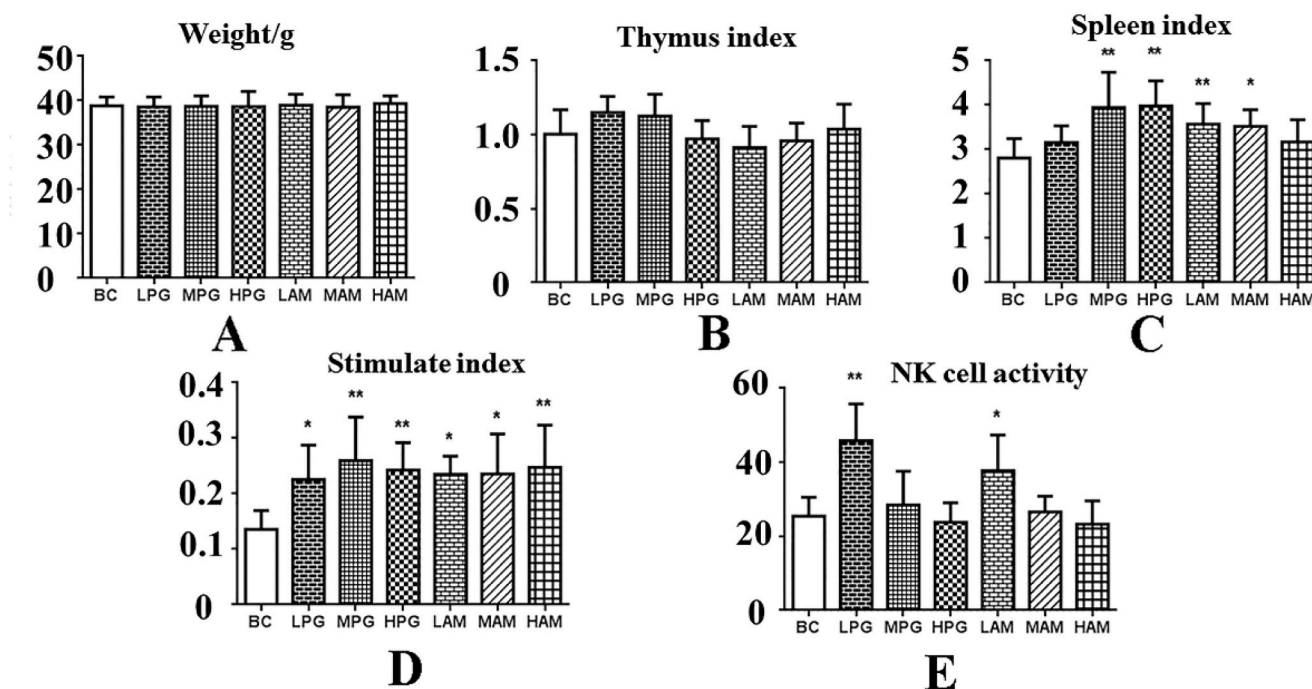


Fig. 3. Pharmacodynamics evaluation of immune index in mice. (A) Weight of BC and administration groups. (B) Thymus index of BC and administration groups. (C) Spleen index of BC and administration groups. (D) Proliferation ability of BC and administration groups. (E) NK cell activity of BC and administration groups. All these results were expressed as mean $\pm$ SD. ** represents $p < 0.01$, * represents $p < 0.05$, compared with the BC group ($n = 12$).

Effect of AM and PG on the cytotoxic activity of NK cells

Compared to BC group, the spontaneous lymphocyte-mediated cytotoxicity of spleen against YAC-1 target cells were increased by 71.74% in LPG ($p < 0.01$) and increased by 48.67% in LAM groups in Fig. 3E. It indicated that AM and PG improved innate immunity in mediating cytotoxic activity of NK cells. The behavior of immune system function is a comprehensive reflection of immune organs and cells *in vivo*. Once physiological equilibrium is broken by internal or external factors, numerous functional and phenotypic changes in monocytes, macrophages, T and B-lymphocyte cells will happen.

In general, spleen and thymus are two important immune organs *in vivo*. Spleen is the biggest immune organ with a plenty of lymphocytes and macrophages, which accounting for 25 percent of weight in total lymphatic tissues. As a central immune organ, spleen is responsible for both humoral and cellular immunity (Mebius and Kraal, 2005). Thymus is a place that lymphocytes differentiate, grow and mature (Miller, 1961). The results indicated that both AM and PG could improve immune activities by increasing spleen index rather than thymus index. Spleen is used a reservoir of blood and clears aged erythrocytes to improve the blood circulation. In this way, blood may play an important role in regulating immune system. During the early phases of infections, innate immune response is triggered prior to adaptive immune response. NK cells are quickly gained from blood circulation and fight with target cells such as tumor cells (Zitti et al., 2017), virus-infected cells (Beltran and Lopez-Verges, 2014) and parasite (Guan et al., 2007). In this way, down-stream adaptive immune response is strengthened (Marcenaro et al., 2006). From the data of NK cell activity assay, the NK cell activity increased in both AM and PG groups with significant difference. In splenocyte proliferation assay, splenocyte proliferation also increased. Long-term administration of AM and PG had a demonstrable positive effect on innate and adaptive immune. Our work was in compliance with previous studies, which AM and PG could improve immune system in normal and disease mode (Shin et al., 2006; Zhao et al., 1990).

All these corresponding data to Fig. 3 of pharmacodynamics evaluation of immune index were given in Table 2.

Screening and discovery the biomarkers of AM and PG

To explore the difference of immune response between them in metabonomic level, we conducted an integrated metabonomic investigation of mice plasma by LC-MS/MS method. A global analysis of metabonomic is untargeted discovery of AM- and PG-induced changes in the diversity of small molecules or endogenous substance in plasma samples. All detected compounds (known or unknown) in plasma samples were taken into investigation by LC-MS method. The typical total ion current chromatograms of QC samples containing AM and PG groups

Table 2
Pharmacodynamics evaluation of immune index in mice ($*p < 0.05$, $**p < 0.01$).

Group	Weight/g	Spleen index (mg/g)	Thymus index (mg/g)	Proliferation ability	NK cell activity (100%)
BC	38.87 ± 1.99	2.81 ± 0.44	1.01 ± 0.15	0.14 ± 0.03	25.41 ± 5.10
LPG	38.54 ± 2.30	3.16 ± 0.37	1.15 ± 0.11	0.23 ± 0.06*	43.65 ± 9.38**
MPG	38.67 ± 2.38	3.95 ± 0.79**	1.13 ± 0.15	0.26 ± 0.08**	28.47 ± 7.08
HPG	38.63 ± 3.43	3.98 ± 0.57**	0.97 ± 0.12	0.24 ± 0.05**	23.80 ± 5.21
LAM	38.94 ± 2.49	3.57 ± 0.47**	0.91 ± 0.14	0.24 ± 0.03*	37.78 ± 8.35*
MAM	38.52 ± 2.82	3.52 ± 0.38*	0.96 ± 0.12	0.24 ± 0.07*	26.57 ± 3.12
HAM	39.35 ± 1.71	3.17 ± 0.50	1.04 ± 0.17	0.25 ± 0.08**	23.26 ± 4.24

were shown in Fig. 4. The data (2176 variables) were used for multivariate statistical analysis. The normalized metabonomic data obtained from UPLC-Q-TOF MS/MS profile data (acquired in positive-ion and negative-ion modes) was processed by MPP software and imported into SIMCA-P software (Version 14.1, Umetrics AB, Sweden) for multivariate data analysis. Unsupervised PCA and supervised OPLS-DA models were made based on LC-MS-acquired data to clarify the metabolic changes and identify the metabolites of BC and treated groups. In positive ion mode, there were 2160 variables in AM groups and 1714 variables in PG groups. In negative ion mode, there were 768 variables in AM groups and 717 variables in PG groups. All those variables were analyzed using PCA model with auto-scaling. AM (LAM, MAM, HAM) groups and BC group were separated into distinct clusters in PCA model in Fig. 5A (positive) and 5C (negative), respectively. There were some partial overlaps among AM groups in negative ion mode. PG (LPG, MPG, HPG) groups and BC group were also separated into distinct clusters in PCA model in Fig. 5E (positive) and 5G (negative), respectively. But there were overlaps to a large extent between MPG and HPG groups. It presented good similarity of various rats in the same group and good separation between BC and treated groups according to the information of PCA scores plot.

Untargeted metabolomic profiling of mice plasma

Metabolic profiling of plasma in treated mice was much different from those of the BC mice. To reduce the individual differences and make a further investigation into those metabolites, which were much close to the separation between the treated and BC groups. OPLS-DA models were built and applied to identify potential biomarkers on the base of the principal components PC1, PC2 and PC3. Score plot represented clear separation of treated and BC groups in Fig. 5B, D, F and H. The validation plot Fig. 5I, J, K and L with 200-iteration permutation test indicated that OPLS-DA models were valid, and the permuted values were significantly lower than un-permuted original values. The potential biomarkers between treated and BC groups were screened by VIP values (> 1) of OPLS-DA model, together with value of $p < 0.05$ and fold change ≥ 2.0 . Endogenous metabolites were found in AM groups and in PG groups, compared to BC group. Furthermore, the identification of metabolite was confirmed by fragmentation of previously detected ions by target mass spectrometry as shown in Table 3.

Metabolic pathway analysis

With these identified metabolites, an integrated metabolic pathway analysis was conducted by MetaboAnalyst 3.6. It revealed that three metabolic pathways including glycerophospholipid, pyrimidine and sphingolipid metabolism were associated with AM-induced mice, seven metabolic pathways including phenylalanine, tyrosine and tryptophan biosynthesis, pentose and glucuronate interconversions, tyrosine metabolism, starch and sucrose, sphingolipid, glycerolipid and glycerophospholipid metabolism were associated with PM-induced mice (Table 4 and Fig. 6).

Correlation analysis

Previous studies mainly focused on detection and identification of biomarkers (Yiqun et al., 2017; Zhang et al., 2016). Diversity of identified endogenous substances were regarded as being most closely to experimental model, regulating related signaling pathways. The relation between biomarkers and metabolic pathways was more like the many-to-many logic relationship. But it is hard to explain which endogenous substance response for specific change in the internal environment. In this correlation analysis section, those mentioned difficulties were smoothly solved. Correlations between detected metabolites and four pharmacodynamic indexes was analyzed by a bivariate correlation test. The relation between them was presented by pearson's correlation coefficient. A two-tailed test of significance using SPSS 24.0 software at significance levels of $p \leq 0.05$ and 0.01. The higher value of

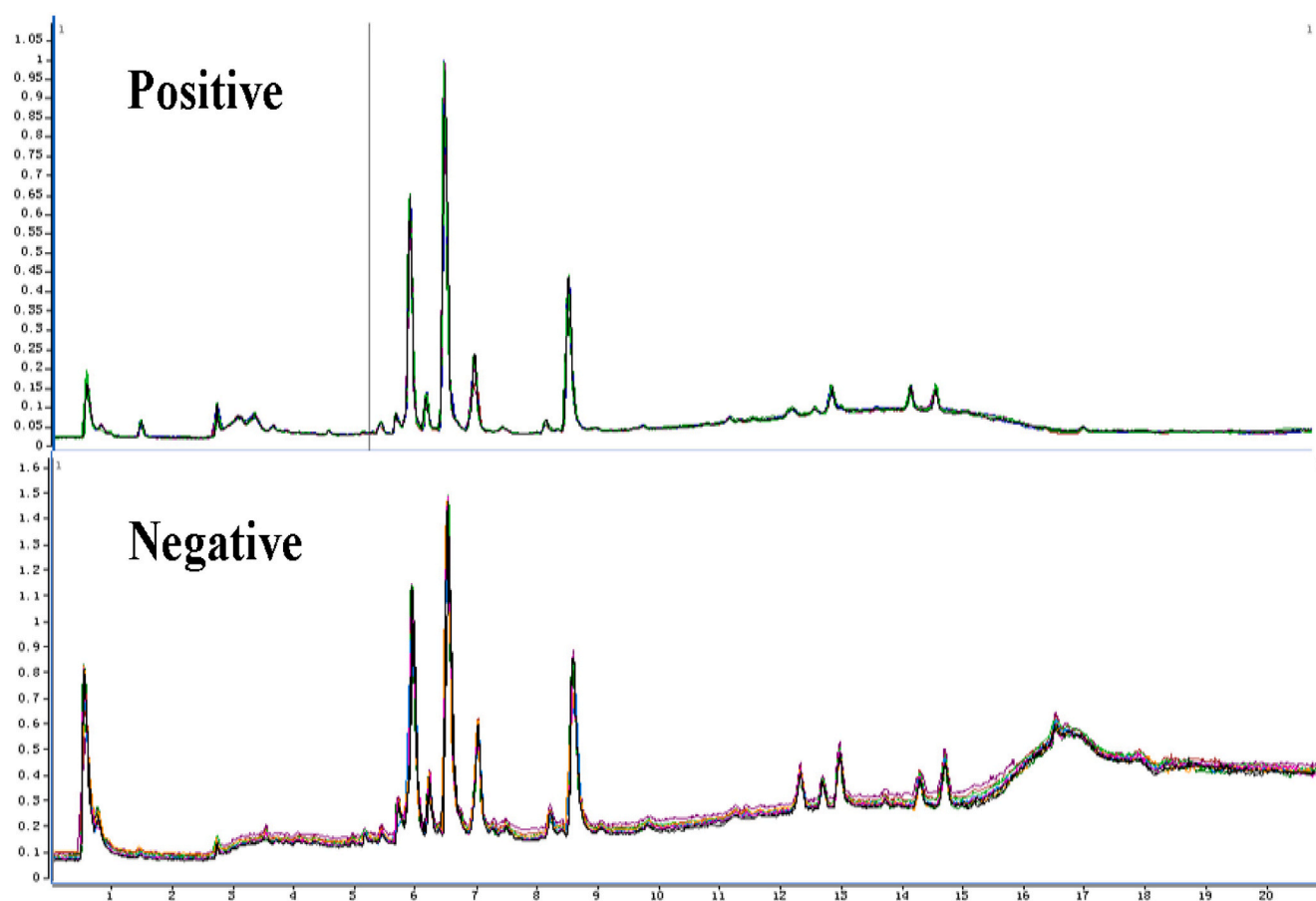


Fig. 4. Typical UPLC/Q-TOF MS total ion current chromatogram of quality control samples in ESI positive and negative ion mode.

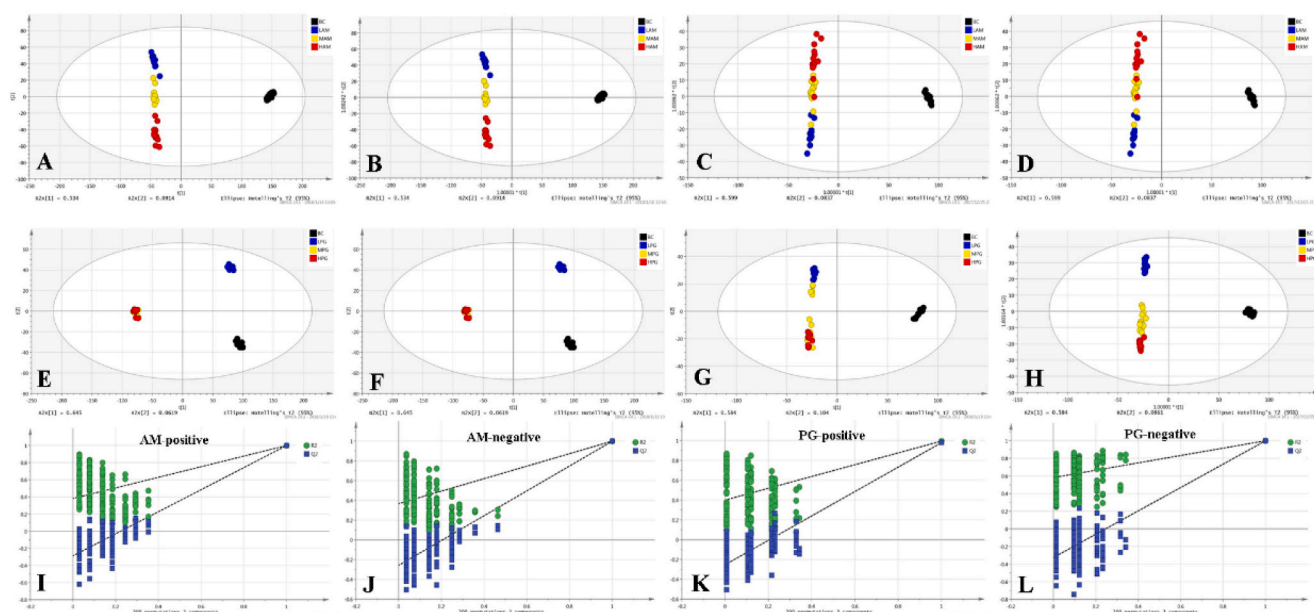


Fig. 5. Plasmatic metabolic-profiling analysis of mice in the AM and PG groups for 30 days. A, C, E and G were principal component analysis (PCA) score plots; B, D, F and H were partial least-squares discriminant analysis (OPLS-DA) score plots; I, J, K and L were repeated permutation.

Pearson's correlation coefficient, the closer relationship between them (Liu et al., 2017). In this way, immune-related metabonomic analysis was deduced (Fig. 7 and Table 5). Glycerophospholipid metabolism was regulated in both AM and PG groups. Pyrimidine metabolism and

sphingolipid metabolism were considered as the special pathway in AM groups. Energy metabolism and glycerolipid metabolism were the special pathways in PG groups.

In AM groups, the significant down regulation of uracil was found,

Table 3

Biomarkers identified in AM and PG groups by UPLC-Q/TOF-MS.

Group	Identify	Rt (min)	m/z (mode)	Formula	VIP	Log FC	Low	Moderate	High
AM	Uracil	0.76	113.0342 (+)	C ₄ H ₄ N ₂ O ₂	1.12	-15.42	↓	↓	↓
	LysoPC (18:3(6Z,9Z,12Z))	4.07	518.3233 (+)	C ₂₆ H ₄₈ NO ₇ P	1.02	12.70	↑	↑	↑
	Desaminotyrosine	3.47	167.1064 (+)	C ₉ H ₁₀ O ₃	1.07	-14.22	↓	—	—
	Pipecolic acid	0.51	130.0868 (+)	C ₆ H ₁₁ NO ₂	1.13	12.88	—	↑	↑
	Octadecanamide	14.57	284.2936 (+)	C ₁₈ H ₃₇ NO	1.09	14.62	—	↑	↑
	Cholesteryl acetate	17.03	429.3727 (+)	C ₂₉ H ₄₈ O ₂	1.17	16.53	—	↑	↑
	Sphinganine	4.34	302.3056 (+)	C ₁₈ H ₃₉ NO ₂	1.42	14.84	—	—	↑
	Eicosadienoic acid	15.15	309.2789 (+)	C ₂₀ H ₃₆ O ₂	1.05	8.26	—	—	↑
	Eicosenoic acid	14.41	333.2794 (+)	C ₂₀ H ₃₈ O ₂	1.10	13.92	—	—	↑
	Indolelactic acid	3.38	204.0665 (-)	C ₁₁ H ₁₁ NO ₃	1.09	14.56	↑	↑	↑
	Gluconic acid	0.58	195.0506 (-)	C ₆ H ₁₂ O ₇	1.07	-13.71	↓	↓	↓
	Pseudouridine	0.64	243.0619 (-)	C ₉ H ₁₂ N ₂ O ₆	1.02	13.47	↑	↑	—
	12,13-DHOME	5.55	313.2383 (-)	C ₁₈ H ₃₄ O ₄	1.05	12.09	↑	↑	↑
	5-HETE	8.19	319.2271 (-)	C ₂₀ H ₃₂ O ₃	1.18	17.80	↑	↑	—
	Oleic acid	14.66	281.2487 (-)	C ₁₈ H ₃₄ O ₂	1.15	12.82	—	—	↑
PG	Purine	6.42	121.0509 (+)	C ₅ H ₄ N ₄	1.63	10.61	—	↑	—
	Indoxyl	0.55	156.0418 (+)	C ₈ H ₇ NO	1.08	-14.96	—	↓	↓
	9-cis-Retinoic acid	7.11	301.2159 (+)	C ₂₀ H ₂₈ O ₂	1.20	-2.02	—	↓	—
	Sphinganine 1-phosphate	5.17	382.2724 (+)	C ₁₈ H ₄₀ NO ₅ P	1.03	13.89	—	↑	—
	PC (14:0/18:0)	14.62	734.5692 (+)	C ₄₀ H ₈₀ NO ₈ P	1.02	-15.44	—	↓	—
	PC (18:0/22:6)	14.24	856.5823 (+)	C ₄₈ H ₈₄ NO ₈ P	1.08	14.88	—	↑	—
	L-Alloisoleucine	0.926	130.0869 (-)	C ₆ H ₁₃ NO ₂	1.17	-16.21	↓	↓	↓
	L-Tyrosine	0.799	180.0672 (-)	C ₉ H ₁₁ NO ₃	1.15	-15.83	↓	↓	↓
	12,13-DHOME	5.54	313.2386 (-)	C ₁₈ H ₃₄ O ₄	1.22	12.35	↑	—	—
	LPA (0:0/16:0)	6.46	409.2362 (-)	C ₁₉ H ₃₉ O ₇ P	1.09	14.57	↑	—	—
	Tauroursodeoxycholic acid	4.20	498.2891 (-)	C ₂₆ H ₄₅ NO ₆ S	1.45	11.00	↑	—	—
	Hippuric acid	3.06	178.0512 (-)	C ₉ H ₉ NO ₃	1.08	13.78	↑	↑	—
	Gluconic acid	0.59	195.0526 (-)	C ₆ H ₁₂ O ₇	1.07	-13.71	↓	↓	—
	Myristic acid	11.62	227.2024 (-)	C ₁₄ H ₂₈ O ₂	1.09	-14.12	↓	↓	—
	LysoPC (14:0)	5.702	466.2966 (-)	C ₂₂ H ₄₆ NO ₇ P	1.05	12.85	—	↑	—
	UDP-glucuronate	5.25	579.0261 (-)	C ₃₀ H ₄₈ O ₁₁	1.09	-14.05	—	↓	—
	LPA (0:0/16:0)	6.51	409.2358 (-)	C ₁₉ H ₃₉ O ₇ P	1.09	14.06	—	↑	↑
	Oleic acid	14.65	281.2487 (-)	C ₁₈ H ₃₄ O ₂	1.37	12.82	—	—	↑

and it was negatively related closely to thymus index, indicating that uracil plays an important in cell proliferation in mice thymus. Uracil is a common naturally occurring pyrimidine found in RNA and an essential key intermediate in adaptive immunity (Doseeth et al., 2011). But, LysoPC(18:3(6Z,9Z,12Z)) and sphinganine may had an opposite effect on thymus, comparing to uracil. Meanwhile, LysoPC(18:3(6Z,9Z,12Z)) also had a positive effect on the proliferation of splenic lymphocytes and enhanced immunity by up regulation of LysoPC(18:3(6Z,9Z,12Z)). LysoPC(18:3(6Z,9Z,12Z)) is a lysophospholipid which have growth-related and -unrelated effects on lymphocytes and macrophages (Gräler and Goetzl, 2002). Besides, enhancement of spleen index was found in both LAM and MAM groups. In PG groups, much difference was found as compared to these in AM groups. Spleen index was enhanced by down regulation of UDP-glucuronate and up regulation of LPA (0:0/16:0). LysoPA(0:0/16:0) is a lysophosphatidic acid and mainly converted from lysophospholipids. Cytotoxic activity of NK cells and proliferation of splenic lymphocytes were found to have strong correlation with PC (14:0/18:0) *in vivo* and correlation coefficients were -0.98 and 0.88, respectively.

Integrated molecular mechanism analysis

With the theoretical data, about 216 pathways with close relationship with AM were deduced and 207 relative pathways were found in PM. All these pathways involved plenty of pharmacologic properties that were controlled by numerous effective substances, leading to final clinical effects. After the pharmacodynamics evaluation in Section 3.4 and correlation analysis in Section 3.7, parts of potential immune-related pathways were found and validated, indicating that AM and PG indeed played a role in regulating immune system. Meanwhile, the bioactive compounds and its targets were also found for the first time. Finally, molecular mechanism of AM and PG in regulating immune system were illustrated in Figs. 8 and 9. In AM, quercetin, hederagenin,

isorhamnetin, 3,9-di-O-methylnissolin, astrapterocarpan, formononetin, kaempferol, betaine, syringaresinol and alpha-L-rhamnose could up-regulate the expression of ACHE and make a positive effect on LysoPC (18:3(6Z,9Z,12Z)), regulating glycerophospholipid metabolism. SPHK1 binding with astragaloside III_{qt} and GLB1 binding with astragaloside I, astragaloside I_{qt}, astragaloside II_{qt}, 5'-hydroxyisomuronulatol-2',5'-di-O-glucoside, astragaloside III_{qt} and astragaloside IV_{qt} were these two key enzymes in regulating sphingolipid metabolism. Meanwhile, astragaloside I_{qt}, astragaloside II_{qt} was also found to have the ability to bind with CDA and decreased the level of exogenous uracil, leading to the change of pyrimidine metabolism.

In PG, 24 ginsenosides such as ginsenoside Ra1, ginsenoside Ra3, ginsenoside Rb1, ginsenoside Rb2 and so on, could bind with the sites of action of UDP-glucuronosyltransferase 2B7 (UGT2B7). This protein together with aldose reductase (AKR1B1, binding molecule: stigmasterol) finally regulate the pentose and glucuronate interconversions. Besides, AKR1B1 was also a key enzyme in glycerolipid metabolism. Choline O-acetyltransferase (CHAT) binding with ginsenoside Ro, ginsenoside Ra0, ginsenoside Ra1, ginsenoside Ra2 and some other ten compounds was an important enzyme in regulating glycerophospholipid metabolism. ACHE was also involved in this metabolism system and played an equal pharmacological function. The binding compounds were beta-sitosterol, kaempferol, stigmasterol, fumarine, arachidonate, frutinone A, girinimbine, ginsenoside D and beta-D-Mannose. Starch and sucrose metabolism were regulated by lysosomal alpha-glucosidase (GAA) and glycogen phosphorylase (PYGM). And the common bioactive compounds were ginsenoside Rh4, 6'-malonylginsenoside Rd1_{qt}, malonylginsenoside Rc_{qt} and ginsenoside Rg3_{qt} (qt1 meant that hydrolysis products). Besides, ginsenoside Rh1 and nonacosanediol-6,8 were found in GAA, and in PYGM were ginsenoside Ro and beta-D-Mannose. All these compounds with the name of label “qt1” meant that the hydrolysis product and could be seen in Table 1.

Table 4
Metaboanalyst analysis of biomarkers in plasma identified from AM and PG groups.

Group	Pathway name	Total	Expected	Hits	Raw p	-log(p)	Impact
LAM	Glycerophospholipid metabolism	30	0.21171	1	0.1932	1.644	0.04444
LAM	Pyrimidine metabolism	41	0.28934	1	0.25514	1.3659	0.04182
LAM	beta-Alanine metabolism	17	0.11997	1	0.11404	2.1712	0
LAM	Arachidonic acid metabolism	36	0.25406	1	0.22754	1.4804	0
LAM	Pantothenate and CoA biosynthesis	15	0.10586	1	0.10126	2.2901	0
MAM	Glycerophospholipid metabolism	30	0.23289	1	0.2104	1.5587	0.04444
MAM	Pyrimidine metabolism	41	0.31828	1	0.27685	1.2843	0.04182
MAM	Lysine degradation	23	0.17855	1	0.16528	1.8001	0
MAM	Steroid biosynthesis	35	0.2717	1	0.24126	1.4219	0
MAM	Pantothenate and CoA biosynthesis	15	0.11644	1	0.11084	2.1996	0
MAM	Arachidonic acid metabolism	36	0.27946	1	0.2473	1.3971	0
MAM	beta-Alanine metabolism	17	0.13197	1	0.12475	2.0815	0
HAM	Sphingolipid metabolism	21	0.20748	1	0.18942	1.6638	0.14286
HAM	Glycerophospholipid metabolism	30	0.2964	1	0.25991	1.3474	0.04444
HAM	Pyrimidine metabolism	41	0.40508	1	0.33833	1.0837	0.04182
HAM	Fatty acid biosynthesis	43	0.42484	1	0.35173	1.0449	0
HAM	Lysine degradation	23	0.22724	1	0.2056	1.5818	0
HAM	Steroid biosynthesis	35	0.3458	1	0.29657	1.2155	0
HAM	Pantothenate and CoA biosynthesis	15	0.1482	1	0.13902	1.9732	0
HAM	beta-Alanine metabolism	17	0.16796	1	0.15613	1.857	0
HAM	Biosynthesis of unsaturated fatty acids	42	0.41496	3	0.007024	4.9584	0
LPG	Phenylalanine, tyrosine and tryptophan biosynthesis	4	0.028229	1	0.027961	3.577	0.5
LPG	Glycerophospholipid metabolism	30	0.21171	1	0.1932	1.644	0.19753
LPG	Tyrosine metabolism	44	0.31052	1	0.27128	1.3046	0.14045
LPG	Glycerolipid metabolism	18	0.12703	1	0.12037	2.1172	0.0192
LPG	Fatty acid biosynthesis	43	0.30346	1	0.26593	1.3245	0
LPG	Ubiquinone and other terpenoid-quinone biosynthesis	3	0.021171	1	0.021037	3.8615	0
LPG	Phenylalanine metabolism	11	0.077629	1	0.075203	2.5876	0
LPG	Aminoacyl-tRNA biosynthesis	69	0.48694	1	0.39397	0.93147	0
MPG	Phenylalanine, tyrosine and tryptophan biosynthesis	4	0.036697	1	0.036233	3.3178	0.5
MPG	Glycerophospholipid metabolism	30	0.27523	3	0.0021248	6.1541	0.38086
MPG	Pentose and glucuronate interconversions	16	0.14679	1	0.13779	1.982	0.2
MPG	Tyrosine metabolism	44	0.40367	1	0.33757	1.086	0.14045
MPG	Starch and sucrose metabolism	19	0.17431	1	0.16159	1.8227	0.03958
MPG	Sphingolipid metabolism	21	0.19266	1	0.17711	1.731	0.02256
MPG	Glycerolipid metabolism	18	0.16514	1	0.15372	1.8726	0.0192
MPG	Phenylalanine metabolism	11	0.10092	1	0.09674	2.3357	0
MPG	Arachidonic acid metabolism	36	0.33028	1	0.28537	1.254	0
MPG	alpha-Linolenic acid metabolism	9	0.082569	1	0.07982	2.528	0
MPG	Pentose and glucuronate interconversions	16	0.14679	1	0.13779	1.982	0
MPG	Fatty acid biosynthesis	43	0.3945	1	0.33125	1.1049	0
MPG	Ubiquinone and other terpenoid-quinone biosynthesis	3	0.027523	1	0.02729	3.6012	0
MPG	Linoleic acid metabolism	6	0.055046	1	0.053892	2.9208	0
MPG	Aminoacyl-tRNA biosynthesis	69	0.63303	1	0.47889	0.73628	0
HPG	Phenylalanine, tyrosine and tryptophan biosynthesis	4	0.01976	1	0.019635	3.9305	0.5
HPG	Glycerophospholipid metabolism	30	0.1482	1	0.13939	1.9705	0.19753
HPG	Tyrosine metabolism	44	0.21736	1	0.19851	1.6169	0.14045
HPG	Glycerolipid metabolism	18	0.08892	1	0.085777	2.456	0.0192
HPG	Biosynthesis of unsaturated fatty acids	42	0.20748	1	0.19028	1.6592	0
HPG	Fatty acid biosynthesis	43	0.21242	1	0.19441	1.6378	0
HPG	Aminoacyl-tRNA biosynthesis	69	0.34086	1	0.29545	1.2192	0
HPG	Ubiquinone and other terpenoid-quinone biosynthesis	3	0.01482	1	0.014757	4.216	0
HPG	Phenylalanine metabolism	11	0.05434	1	0.053201	2.9337	0

Discussion

The immune system is a huge defensive project that protect our body from infectious organisms and other invaders. The defensive project is made up of a complex network, which contains immune cells, tissues, and organs. With the prevalence of immune-related diseases, powerful treatment strategies to these complex systemic diseases are immediately needed. TCM have the characteristics of numerous kinds of potential active compounds which produce synergistic effect on biological diseases by acting on the multi-targets and multi-pathways (Ma et al., 2013). In this way, multiple-targets therapy put a good treatment on complex disease systems and ultimately lead to the therapeutic effect (Zimmermann et al., 2007). In TCM practice, multiple drugs are usually used together to get a good therapeutic effect, such as AM and PG. Even though these two drugs are typical drug partner and have a clear clinical application, but there is still a need to find out underlying synergistic action mechanisms between them. There is a common understanding

that the complex biological interactions among the multiple bioactive compounds from the multi-herbs has effective synergistic therapeutic effects (Berenbaum, 1981; Zhou et al., 2016). However, the underlying molecular mechanism of these synergistic effects remains is still elusive due to several reasons, including the incomplete understanding of bioactive compounds, the ambiguous biological function of potential related targets and pathways (Polya, 2003). Thus, it is extremely difficult or impracticable to make a clear explanation of action mechanisms by traditional methods. All these existing difficulties or problems presents us challenges and need urgent solutions. Here, a system pharmacology method is established and applied to carry on a deep insight into the comparative analysis of molecular mechanism of AM and PG in regulating immune system.

AM and PG are widely used for improving the immunity of human beings in Chinese long history clinical practice with higher dosage and longer time usage. Thus, a matched full-scale method is needed to monitor the complex biochemical reaction *in vivo*. In this study,

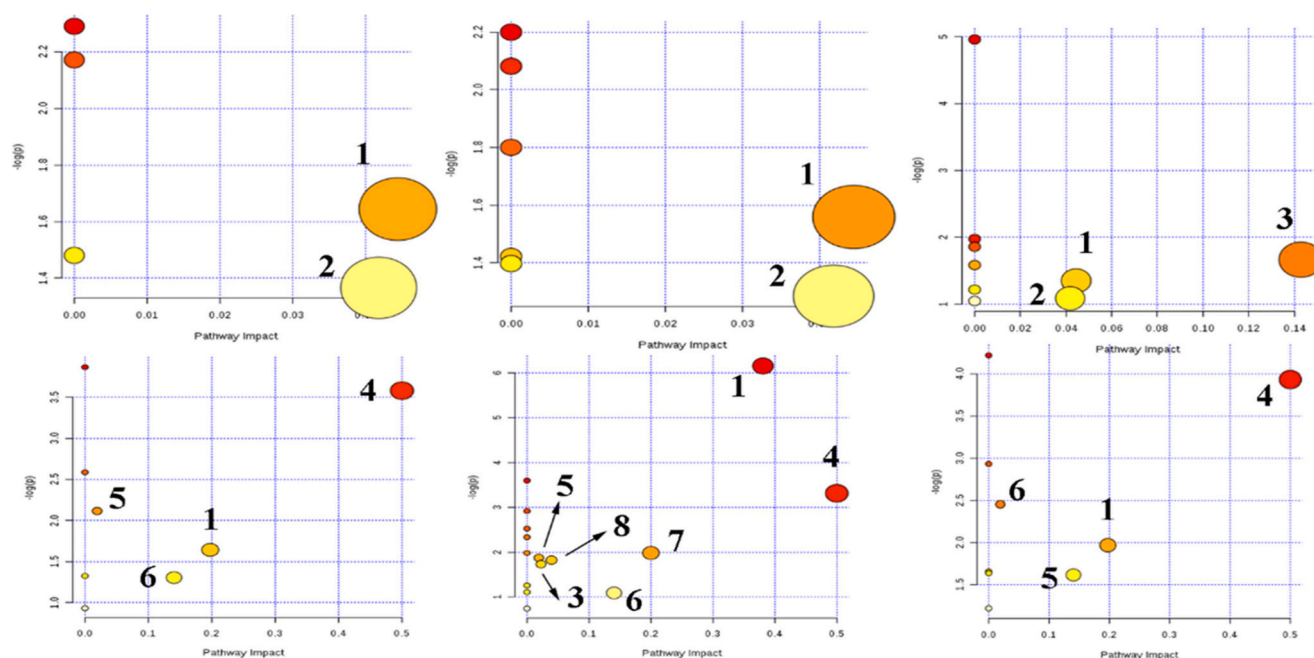


Fig. 6. Summary of pathway analysis of biomarkers in plasma identified from AM and PG groups. Key:1, Glycerophospholipid metabolism; 2, Pyrimidine metabolism; 3, Sphingolipid metabolism; 4, Phenylalanine, tyrosine and tryptophan biosynthesis; 5, Glycerolipid metabolism; 6, Tyrosine metabolism; 7, Pentose and glucuronate interconversions; 8, Starch and sucrose metabolism.

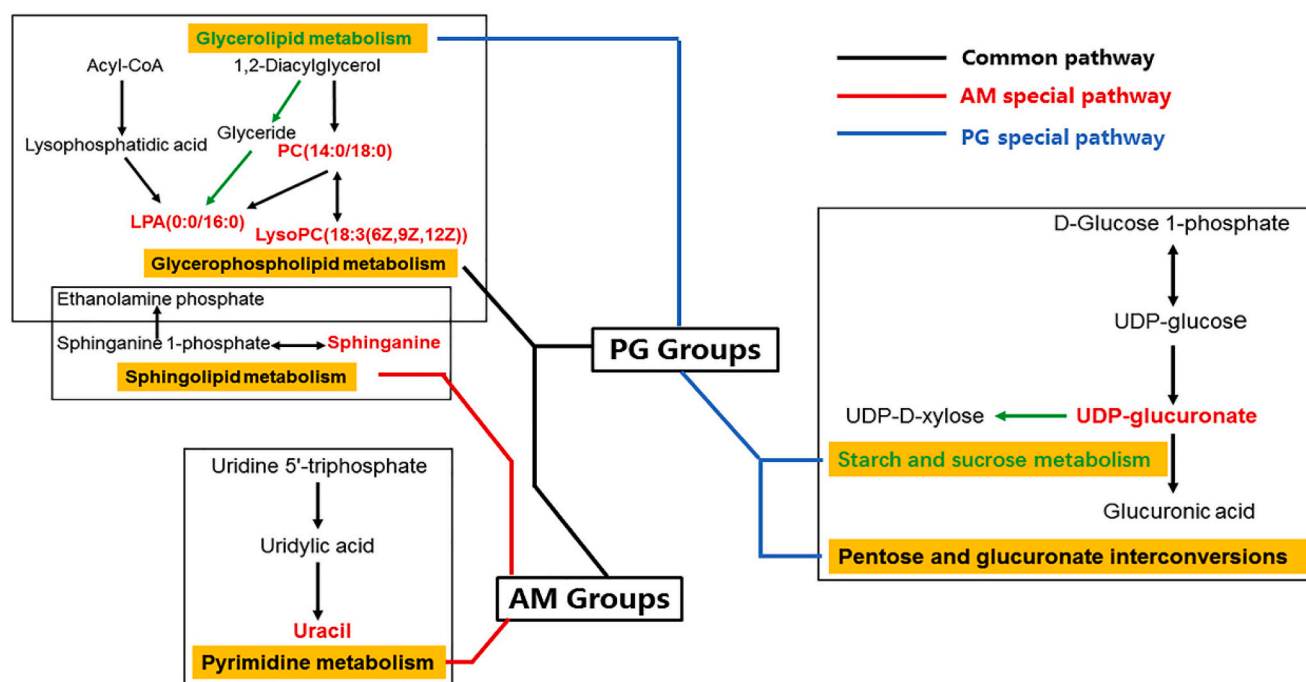


Fig. 7. The similarities and differences of immune-related pathways analysis between AM and PG groups.

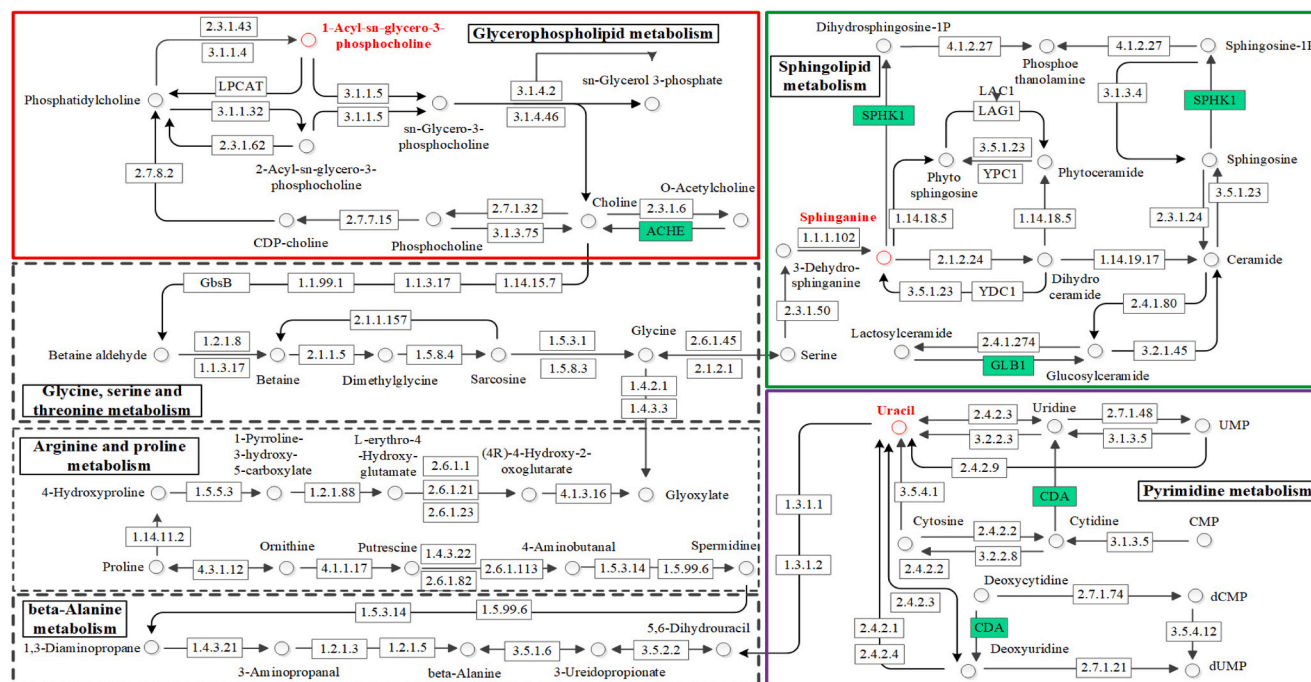
bioactive compounds, potential targets and its related pathways were fully analyzed by the TCMSP analysis platform. In this efficient platform, ADME evaluation system, WES drug targeting model and GO enrichment analysis provide us with a clear theoretical molecular mechanism of these two tonic herbs in regulating immune system, contributing to a better understanding of interaction between molecules and targets. And then, experimental validation is conducted on the base of a comprehensive animal experiment. In these validation processes, conventional pharmacological method and metabolomic analysis are simultaneously

performed, and the results obtained from these experiments are cross-over analyzed together. In pharmacological section, immunological indexes are used to assess the immune regulation of experimental model in cellular and organ level with macroscopic scale. And then, experiment on metabolomics brings out an efficient strategy in detecting, identifying endogenous biomarkers and analyzing the immune regulation profiling in molecular level with microscopic scale.

From computational to biological models, from microscopic to macroscopic scale, a comprehensive and integrated approach was

Table 5List of correlation between biomarkers and pharmacodynamic indexes (** represents $p < 0.01$, * represents $p < 0.05$).

Group	Name	Rt (min)	m/z	Trend	Pearson Correlation Coefficient / Pharmacodynamic indexes	spleen Index	thymus index	cell proliferation rate	cytotoxic activity
LAM	Uracil	0.76	(+) 113.0342	↓	–	–	–0.63* / 0.91	–	–
LAM	LysoPC(18:3(6Z,9Z,12Z))	4.07	(+) 518.3233	↑	+0.66* / 3.57**	–	–	–	–
MAM	LysoPC(18:3(6Z,9Z,12Z))	4.06	(+) 518.3240	↑	+0.60* / 3.52*	–	–	–	–
HAM	LysoPC(18:3(6Z,9Z,12Z))	4.06	(+) 518.3238	↑	–	–	+0.54* / 1.04	+0.55* / 0.25**	–
HAM	Sphinganine	4.34	(+) 302.3056	↑	–	–	+0.61* / 1.04	–	–
LPG	LPA(0:0/16:0)	6.46	(-) 409.2362	↑	–	–	–	–	+0.66* / 71.7**
MPG	UDP-glucuronate	5.25	(-) 579.0261	↓	–0.64* / 3.95**	–	–	–	–
MPG	PC(14:0/18:0)	14.63	(+) 734.5692	↓	–	–	–0.98** / 0.26*	–	+0.88* / 28.5
HPG	LPA(0:0/16:0)	6.51	(-) 409.2356	↑	+0.71* / 3.98**	–	–	–	–

**Fig. 8.** The target proteins profiling reveals the molecular mechanism of AM with immune-related pathways. The red words represent biomarkers detected in experiment validation and the green labels represent target proteins.

gradually building up and a solid explanation of the molecular mechanism of immune regulation was obtained. The biological processes that lead to immune diseases, which initiated by AM and PG showed that glycerophospholipid metabolism was the common gene-pathway of these tonic drugs. It was reported that compounds of glycerophospholipid are important constituents of membranes of mammalian cells (Hermansson et al., 2011) and these compounds have the ability to activate varieties of immune cells such as monocytes and neutrophils (Nishioka et al., 1998). LysoPC (18:3(6Z,9Z,12Z)) (KEGG: C04230) is a compound of glycerophospholipid and is found in internal environment as a biomarker by metabonomic method. It is an intermediate in glycerophospholipid metabolism and it is the precursor of choline. Choline is a key synthetic material for the cell membranes (Zeisel, 2000), which protect immune system from microbial pathogens and autoimmune disease, and also play an essential role in wound healing and tissue remodeling (Silverman et al., 2009). Besides, choline is also involved in the development of cholinergic neurotransmission and lipid metabolism (Zeisel, 2000). In glycerophospholipid metabolism, acetylcholine is another key compound in transmission of neural signal and the expression of acetylcholine in lymphocytes could regulate of immune function (Kawashima and Fujii, 2004). Acetylcholinesterase and choline O-acetyltransferase are the two key catalytic enzymes that make mutual transformation between choline and acetylcholine. Luckily, these two

enzymes were instantaneously targeted got by computerized algorithm and biological process analysis in our experiment. After analysis of common pathway, characteristic regulation pathways were also obtained, and pyrimidine metabolism and sphingolipid metabolism were found in AM, glycerolipid metabolism, pentose and glucuronate interconversions, starch and sucrose metabolism were in PG. Pyrimidine compounds are widely distributed in mammalian animals and provide materials for a variety of mammalian metabolism throughout life. The promotion of cell proliferation is the main function of pyrimidine metabolism (Marijnen et al., 1989). Uracil is a base of DNA replication (Sire et al., 2008), playing an important role in protecting host from being attacked by pathogenic microorganisms (Maul and Gearhart, 2014). Sphingolipid metabolism was found to affect immune function by binding CD1 proteins (Huang et al., 2011) and used as chemoattractant for immune cells (Hannun and Obeid, 2008). CD4+ T cell function was typically regulated by sphingolipid metabolites in several ways such as regulation of T cell activation, survival and proliferation (Molano et al., 2012). In glycerolipid metabolism, LPA (16:0/0:0) is a kind of fatty acid which is stored as energy in complex organisms and animals. Glycerolipid metabolism plays a role in maintaining body temperature by modulating glycerolipid fatty acid cycle in vivo. As a regulation mechanism during animal signing processes, it contains numerous formation and hydrolysis of glycerolipid and releases heat with the consumption of

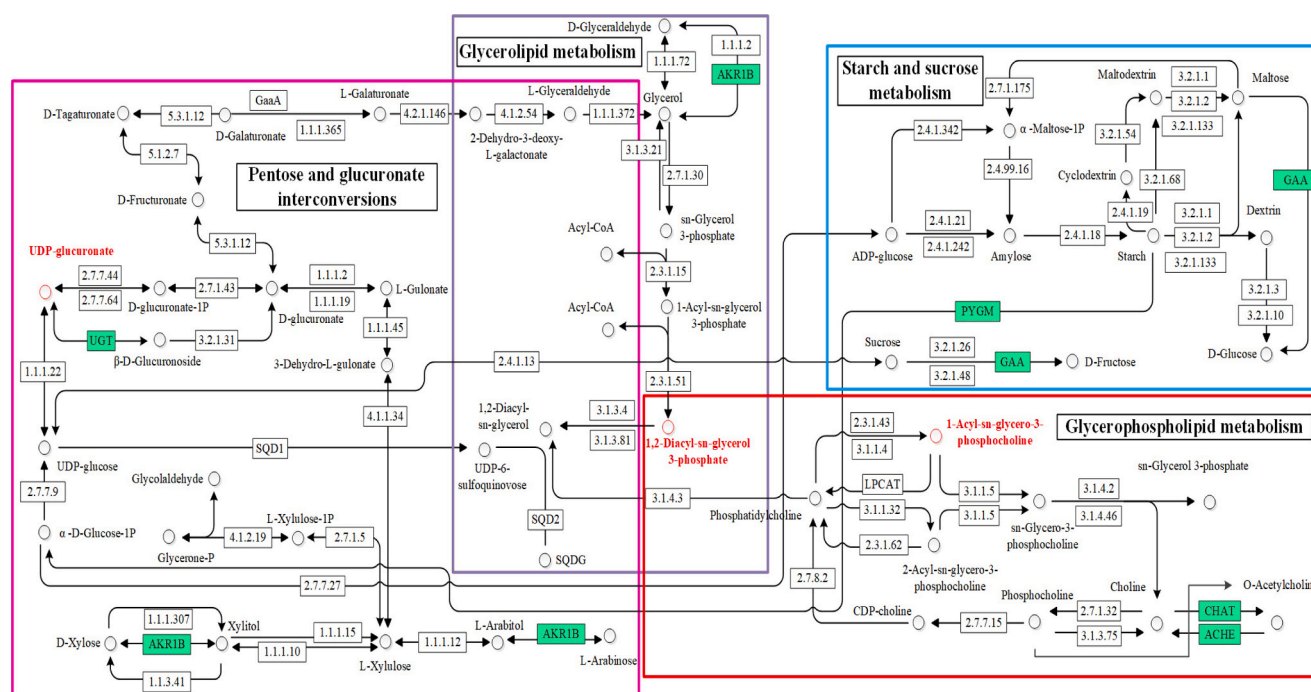


Fig. 9. The target proteins profiling reveals the molecular mechanism of PG with immune-related pathways. The red words represent biomarkers detected in experiment validation and the green labels represent target proteins.

ATP (Prentki and Madiraju, 2008). Pentose and glucuronate interconversions & Starch and sucrose metabolism: pentose and glucuronate provide numerous ribose 5-phosphate for nucleotide and nucleic acid synthetic reactions, playing an important role in cell metabolism (Sochor et al., 1979). UDP-glucose dehydrogenase is an essential process in pentose and glucuronate interconversions and UDP-glucose is converted to UDP-glucuronate (Spicer et al., 1998). When host is attacked, damaged cells will release UDP-glucose which trigger innate mucosal immune responses (Arase et al., 2009). AM and PG not only share the same regulation pattern of glycerophospholipid metabolism, especially in choline esterase, but also present the specific regulating behavior, AM trend to promote cell proliferation and DNA replication and PM emphasis on providing the body with ATP.

Conclusion

With the deep analysis of molecular mechanism of AM and PG in immune regulation, a clear result was obtained that AM and PG had the accurate, broad and comprehensive strategy in regulating human immune system. We speculate that the combination of these two tonic drugs should more effective treatment than the use of single drug in immunosuppression models. That meaningful work could contribute to the interpretation of compatibility of TCM prescriptions. Besides, nervous system was seen to have a close relationship with immune system in this our experiment and many literatures also demonstrated this relationship (Aliev, 1962; Rabin et al., 1988; Sierra Vazquez et al., 1998; Webb and Smith, 1966). So, a deeper insight into enzymatic network around AM and PG will provide us a better evidence of the mechanism of immune regulation, such as reaction site, enzyme activity and its products. In this way, new sight of targets seeking and validation in immune therapy is found and suitable for other systemic disorders.

Conflict of Interest

The authors have no conflicts of interest to declare.

Acknowledgments

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Author contributions

Participated in research design: Junqiu Liu, Yuesheng Wang, Guoyin Kai. Conducted experiments: Junqiu Liu, Guoliang Xu. Performed data analysis: Junqiu Liu, Yuesheng Wang.

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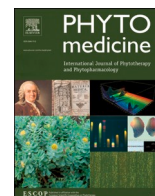
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Corrigendum

Corrigendum to <Systematic exploration of Astragalus membranaceus and Panax ginseng as immune regulators: Insights from the comparative biological and computational analysis>

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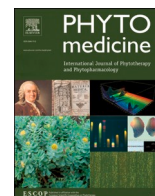
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Corrigendum

Corrigendum to Systematic exploration of Astragalus membranaceus and Panax ginseng as immune regulators: Insights from the comparative biological and computational analysis Phytomedicine 86 (2019) 153077

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